

A Counterexample to Conjecture 00000000013

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Statement

Conjecture 00000000013 claims that for every integer $n \geq 3$ there is a prime p in the half-open interval

$$2^n \leq p < 2^{n+1}$$

whose binary expansion contains exactly three 1's.

We disprove the conjecture by taking $n = 8$.

The counterexample

Every integer in $[2^8, 2^9)$ with exactly three 1's in its binary expansion has the form

$$2^8 + 2^a + 2^b, \quad 0 \leq b < a < 8.$$

There are $\binom{8}{2} = 28$ such numbers. They are listed below, together with a proper divisor. Thus every one is composite.

n	divisor	n	divisor	n	divisor	n	divisor
259	7	261	3	262	2	265	5
266	2	268	2	273	3	274	2
276	2	280	2	289	17	290	2
292	2	296	2	304	2	321	3
322	2	324	2	328	2	336	2
352	2	385	5	386	2	388	2
392	2	400	2	416	2	448	2

For completeness, the corresponding factorizations include

$$\begin{aligned} 259 &= 7 \cdot 37, & 261 &= 3 \cdot 87, & 262 &= 2 \cdot 131, & 265 &= 5 \cdot 53, \\ 266 &= 2 \cdot 133, & 268 &= 2 \cdot 134, & 273 &= 3 \cdot 91, & 274 &= 2 \cdot 137, \\ 276 &= 2 \cdot 138, & 280 &= 2 \cdot 140, & 289 &= 17^2, & 290 &= 2 \cdot 145, \\ 292 &= 2 \cdot 146, & 296 &= 2 \cdot 148, & 304 &= 2 \cdot 152, & 321 &= 3 \cdot 107, \\ 322 &= 2 \cdot 161, & 324 &= 2 \cdot 162, & 328 &= 2 \cdot 164, & 336 &= 2 \cdot 168, \\ 352 &= 2 \cdot 176, & 385 &= 5 \cdot 77, & 386 &= 2 \cdot 193, & 388 &= 2 \cdot 194, \\ 392 &= 2 \cdot 196, & 400 &= 2 \cdot 200, & 416 &= 2 \cdot 208, & 448 &= 2 \cdot 224. \end{aligned}$$

Hence there is no prime in $[2^8, 2^9)$ whose binary expansion has exactly three 1's. Since the conjecture asserts this for every $n \geq 3$, the single counterexample $n = 8$ disproves it.

Formal verification

The accompanying Lean 4 project formalizes the same finite verification. It defines $\text{Composite}(N)$ to mean that there is a divisor d with $1 < d < N$ and $d \mid N$. A finite computation verifies that every pair of exponents $0 \leq b < a < 8$ gives a composite number $2^8 + 2^a + 2^b$. The final Lean theorem derives the negation of the conjecture's assertion at $n = 8$. The project uses Lean 4.33.1 and only its standard library.